PEOPLE'S DEMOCRATIC REPUBLIC OF ALGERIA
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Site-to-Site IPsec VPN

Design and Implementation Report

Module: Network Foundation **Project Type:** CCNA Security Oriented **Academic Year:** 2025–2026

1. INTRODUCTION

In modern enterprise environments, secure communication between geographically distributed networks is a fundamental requirement. Public networks such as the Internet are inherently insecure, exposing transmitted data to various threats including interception, manipulation, and unauthorized access.

This report presents the design and implementation of a Site-to-Site IPsec Virtual Private Network (VPN) using Cisco routers. The main objective is to securely interconnect two remote local area networks (LANs) over an untrusted ISP infrastructure while ensuring confidentiality, integrity, and authentication. The project follows CCNA Security guidelines and reflects real-world enterprise deployment scenarios.

2. THEORETICAL FRAMEWORK

2.1 Virtual Private Network (VPN)

A Virtual Private Network (VPN) is a technology that allows private networks to communicate securely over public infrastructures. VPNs create encrypted tunnels that protect transmitted data from unauthorized access. The key advantages of VPN technology include:

- Secure data exchange over public networks
- Reduced operational costs compared to dedicated leased lines
- High scalability and organizational flexibility

2.2 Internet Protocol Security (IPsec)

IPsec is a suite of protocols designed to secure IP communications at the network layer (Layer 3 of the OSI model). Because IPsec operates independently of applications, it provides transparent security for all IP-based traffic traversing the network.

2.3 IPsec Security Services

- **Confidentiality:** Ensured through symmetric encryption algorithms such as AES.
- **Integrity:** Guaranteed using cryptographic hashing algorithms such as SHA.
- **Peer Authentication:** Confirms identities using pre-shared keys or digital certificates.
- **Anti-Replay Protection:** Prevents malicious packet interception and retransmission.

2.4 IPsec Operating Modes

- **Transport Mode:** Encrypts only the data payload while leaving the original IP header intact.
- **Tunnel Mode:** Encrypts the entire original IP packet and prepends a new IP header. Selected for this project as it is the standard mode for site-to-site VPN deployments.

3. NETWORK TOPOLOGY & ADDRESSING

3.1 Topology & Objectives

The network topology consists of two remote sites connected across a simulated ISP network infrastructure. Each site is equipped with a Cisco router acting as a VPN gateway to protect its internal local network.

- Secure end-to-end inter-site connectivity
- Maintainable configuration compliant with CCNA Security standards
- Future scalability for additional branch integration

3.2 Security License Activation

On Cisco 1941 routers, security feature packages must be enabled prior to configuring IPsec protocols:

LISTING 1: SECURITY LICENSE ACTIVATION

```
      Router(config)# license boot suite Security9k Router(config)# exit Router# write
memory Router# reload
```

3.3 IP Addressing Table

Device	Interface	IP Address	Subnet Mask
Router 1 (WAN)	Gi0/0	209.165.100.1	255.255.255.252
Router 1 (LAN)	Gi0/1	192.168.1.1	255.255.255.0
Router 3 (WAN)	Gi0/0	209.165.200.1	255.255.255.252
Router 3 (LAN)	Gi0/1	192.168.3.1	255.255.255.0

4. TECHNICAL CONFIGURATION

4.1 IKE Phase 1: ISAKMP Policy

IKE Phase 1 establishes a secure control management channel between VPN peers.

LISTING 2: IKE PHASE 1 CONFIGURATION

```
      crypto isakmp policy 10 encr aes 256 hash sha group 5 authentication pre-share
exit crypto isakmp key secretkey address <Peer_WAN_IP>
```

4.2 IKE Phase 2: IPsec Transform Set & ACL

Phase 2 negotiates the encryption and integrity parameters for data traffic.

LISTING 3: TRANSFORM SET & ACCESS CONTROL LIST

```
crypto ipsec transform-set VPN-SET esp-aes 256 esp-sha-hmac mode tunnel access-list 100 permit ip 192.168.1.0 0.0.0.255 192.168.3.0 0.0.0.255
```

4.3 Crypto Map Configuration

LISTING 4: CRYPTO MAP BINDINGS

```
crypto map IPSEC-MAP 10 ipsec-isakmp set peer <Peer_WAN_IP> set transform-set VPN-SET match address 100 interface GigabitEthernet0/0 crypto map IPSEC-MAP
```

5. VERIFICATION, ANALYSIS & MAINTENANCE

5.1 Verification Commands

Establishment of Phase 1 and Phase 2 tunnels is verified through operational status commands:

LISTING 5: VERIFICATION COMMANDS

```
Router# show crypto isakmp sa # Expected State: QM_IDLE Router# show crypto ipsec sa # Verify active packet encapsulation and decapsulation: # pkts encaps: 11, pkts decaps: 11
```

Successful encapsulation and decapsulation statistics confirm functional VPN operation and data path encryption.

5.2 Security Analysis

The implemented IPsec VPN solution successfully guarantees confidentiality, data integrity, and peer authentication. By enforcing AES-256 encryption and SHA hashing within Tunnel Mode, enterprise traffic across public WAN media is effectively shielded against packet sniffing, spoofing, and man-in-the-middle (MitM) vectors.

5.3 Troubleshooting & Diagnostics

- **Phase 1 Failures:** Often caused by mismatched ISAKMP parameters (mismatched encryption, hash, Diffie-Hellman group) or incorrect pre-shared keys.
- **Phase 2 Failures:** Typically result from non-matching Transform Sets or asymmetrical Access Control Lists (ACLs).

LISTING 6: DEBUGGING COMMANDS

```
debug crypto isakmp debug crypto ipsec
```

5.4 Performance Considerations

Cryptographic encapsulation introduces processing overhead on gateway routers. Hardware acceleration capabilities, interface bandwidth limitations, and latency factors should be evaluated when sizing enterprise IPsec deployments.

5.5 Future Enhancements

- Migration to IKEv2 for higher performance and resilience.
- Implementation of Public Key Infrastructure (PKI) and digital certificates.
- Integration with centralized SIEM solutions for audit and monitoring.

6. Conclusion

This project successfully demonstrates the end-to-end design, implementation, and operational validation of a Site-to-Site IPsec VPN using Cisco routers. The configuration complies with CCNA Security guidelines, establishing a robust framework for secure multi-site enterprise interconnectivity.